# Audio Signal Processing Assistant Agent

## Project Report

IBM University Engagement Project Submission

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| **Domain** | Electronics & Communication Engineering |
| **Project** | Audio Signal Processing Assistant using IBM Watsonx.ai & LangFlow |
| **Platform** | IBM watsonx.ai \| IBM Granite 8B \| LangFlow |
| **Status** | Active Development |
| **Report Date** | June 2026 |

## 1. Executive Summary

This report presents the design, development, and implementation of the **Audio Signal Processing Assistant Agent**, an AI-powered conversational system built under the IBM University Engagement Programme. The project leverages **IBM watsonx.ai**, **IBM Granite 8B Code Instruct**, and **LangFlow** to deliver an intelligent, domain-specific assistant that helps students, engineers, and audio enthusiasts understand digital signal processing (DSP) concepts and troubleshoot audio issues through natural language interaction.

The agent combines agentic AI, specialized prompt engineering, and a low-code workflow to simulate the expertise of a virtual audio engineering consultant — providing instant, context-aware responses without requiring users to navigate fragmented technical documentation.

## 2. Problem Statement

Audio signal processing is a fundamental discipline in electronics, communications, and digital engineering. However, learners and practitioners frequently encounter barriers that hinder effective understanding and problem resolution:

| Challenge | Impact |
| --- | --- |
| Conceptual Difficulty | Complex DSP topics — FFT, filtering, sampling, quantization — are hard to grasp without guided, contextual help. |
| Troubleshooting Gaps | Users struggle to diagnose real-world issues such as humming, clipping, distortion, echo, feedback, and signal interference. |
| Time Inefficiency | Extensive searching through scattered documentation significantly delays problem resolution. |
| Knowledge Fragmentation | Relevant information is spread across textbooks, datasheets, and online forums, with no unified interface. |

### Objective

The project aims to build an AI-powered assistant that addresses all of the above by providing:

- **Audio Knowledge Assistance** — Explanations of DSP concepts including filtering, sampling, quantization, FFT, and frequency analysis.
- **Audio Troubleshooting** — Identification and resolution of common audio problems.
- **Intelligent Question Answering** — Technical responses on audio systems, amplifiers, microphones, and signal processing.
- **AI-Powered Analysis** — Context-aware recommendations using IBM Granite foundation models.
- **Educational Support** — Simplified explanations and practical examples for students.

## 3. Technology Stack

| Technology | Role |
| --- | --- |
| LangFlow | Workflow Design & Orchestration — visual, low-code pipeline builder |
| IBM watsonx.ai | Enterprise AI platform providing secure model access and project management |
| IBM Granite 8B Code Instruct | Foundation LLM powering natural language understanding and response generation |
| Prompt Engineering | AI behaviour configuration — domain-specific persona and constraints |
| Agentic AI | Autonomous, context-aware decision-making beyond a simple chatbot |
| NLP | Query understanding and intelligent response generation |

## 4. Proposed Solution

The Audio Signal Processing Assistant Agent is built on a specialised agentic AI framework that functions as a **virtual audio engineering expert**. The system integrates IBM watsonx.ai, IBM Granite 8B, LangFlow for workflow orchestration, and specialised prompt engineering to deliver four distinct AI capabilities:

| Specialised Agent | Capability |
| --- | --- |
| Audio Knowledge Agent | Provides explanations of fundamental and advanced audio signal processing concepts including filtering, sampling, quantization, Fourier Transform, frequency analysis, and digital audio processing. |
| Troubleshooting Agent | Identifies common audio system issues — humming noise, clipping, distortion, feedback, interference, grounding problems — and suggests practical corrective measures. |
| Signal Analysis Agent | Assists users in understanding signal characteristics, filter selection, frequency response, noise reduction techniques, and signal conditioning methods. |
| Learning Support Agent | Supports students and beginners by simplifying complex DSP concepts and providing educational explanations with real-world examples. |

## 5. Architecture & Workflow

### 5.1 LangFlow Component Pipeline

| Component | Function |
| --- | --- |
| Chat Input | Receives and captures user queries in natural language |
| Prompt Template | Defines assistant behaviour — injects audio DSP expert persona, constraints, and user message formatting |

| Component | Function |
| --- | --- |
| IBM watsonx.ai Node | Authenticates and routes API calls to IBM cloud infrastructure |
| IBM Granite Model | Processes queries and generates domain-specific intelligent responses |
| Chat Output | Formats and displays AI-generated responses to the user |

### 5.2 System Architecture Diagram

The figure below shows the full LangFlow workflow as configured in the development environment. The pipeline illustrates the linear data flow from Chat Input through the Prompt Template to IBM watsonx.ai and the IBM Granite model, terminating at Chat Output.

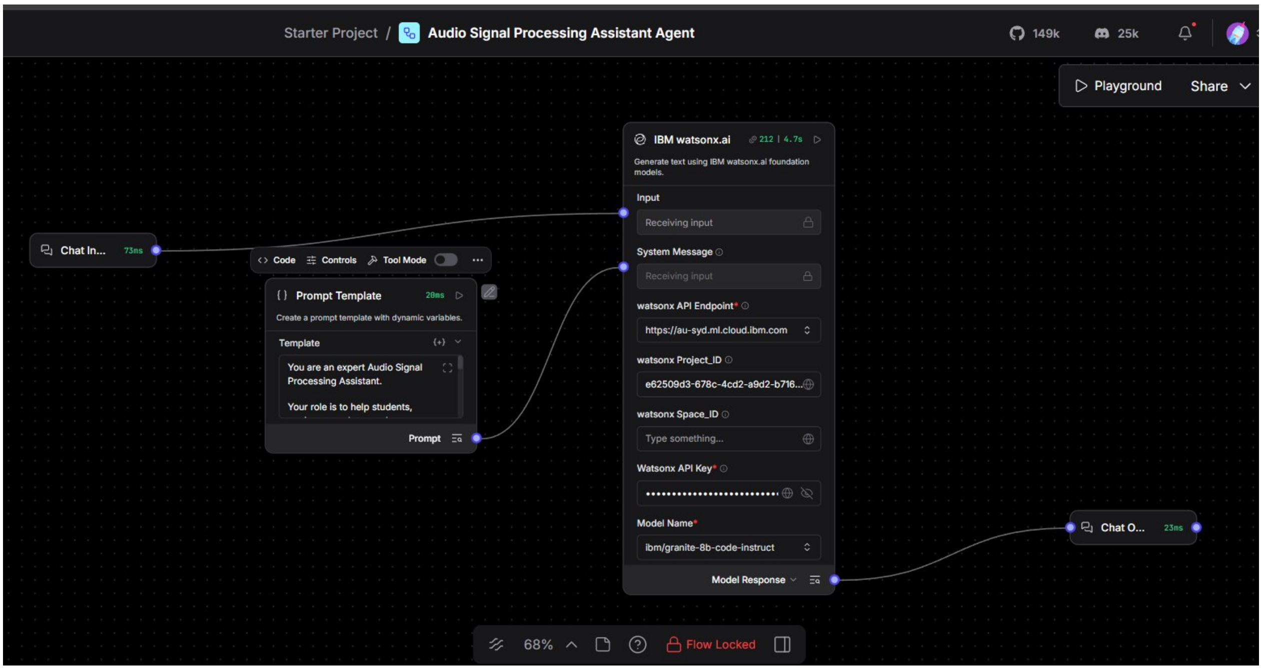


**Figure 1: Complete LangFlow Workflow — Audio Signal Processing Assistant Agent**

The LangFlow canvas shows the Chat Input node (left) connected to the Prompt Template, which feeds into IBM watsonx.ai configured with the IBM Granite 8B Code Instruct model. The Model Response output routes to the Chat Output node (right).

## 6. Role of Agentic AI

Unlike traditional rule-based chatbots, the Audio Signal Processing Assistant Agent functions as an **autonomous intelligent agent**. Using IBM watsonx.ai and IBM Granite foundation models, the system understands user queries, reasons independently about audio-related issues, and generates context-aware recommendations — exhibiting genuine problem-solving behaviour:

- **Intelligent Decision-Making** — The system analyses problems and selects the most appropriate explanation or troubleshooting path.

• **Context-Aware Responses** — Maintains understanding of the specific audio engineering context throughout the conversation.

• **Audio Troubleshooting Support** — Diagnoses issues from symptom descriptions rather than requiring exact technical terminology.

• **Personalised Technical Guidance** — Adapts the complexity of explanations to the user's apparent expertise level.

• **Faster DSP Knowledge Access** — Instant retrieval of audio engineering expertise without manual searching.

• **Enhanced Learning Experience** — Interactive, guided explanations accelerate comprehension for students.

# 7. Agent Response Demonstrations

The following screenshots demonstrate the agent in operation within the LangFlow Playground environment, validating the system's capability to handle real-world audio DSP queries.

## 7.1 Amplifier Noise Troubleshooting

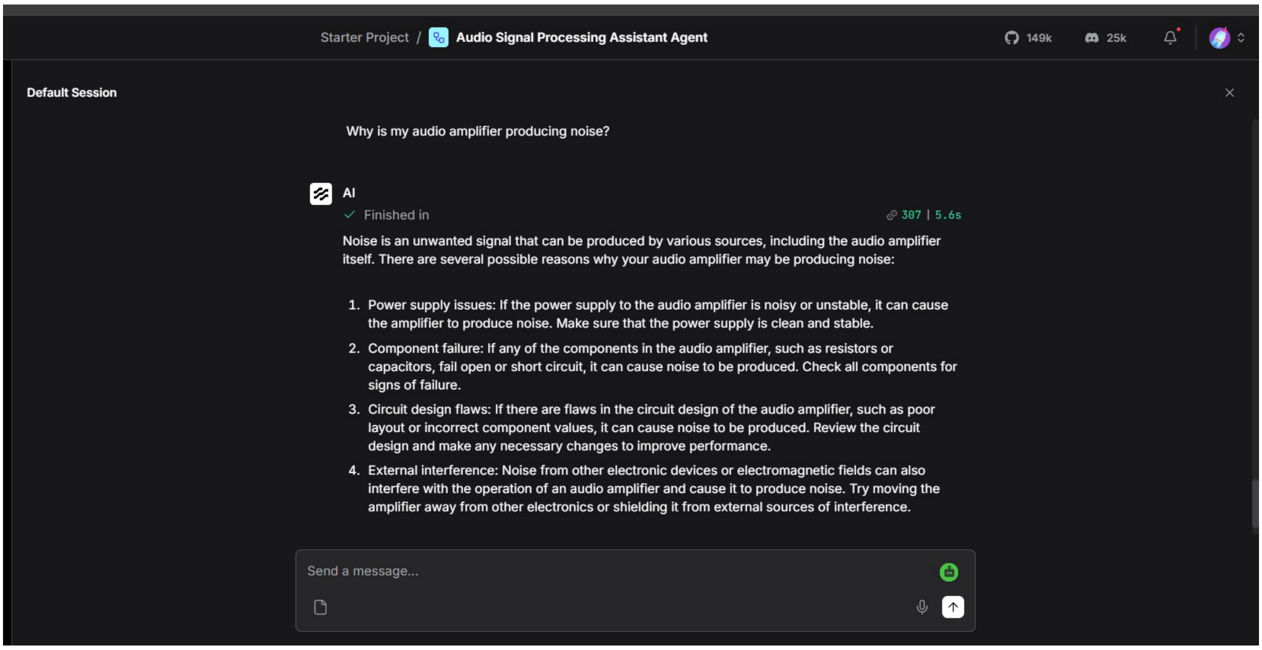


**Figure 2: Agent Response — Audio Amplifier Noise Troubleshooting**

Query: "Why is my audio amplifier producing noise?" The IBM Granite model responded with four structured diagnostic causes — power supply issues, component failure, circuit design flaws, and external electromagnetic interference — each with actionable corrective steps.

## 7.2 Op-Amp Distortion Troubleshooting

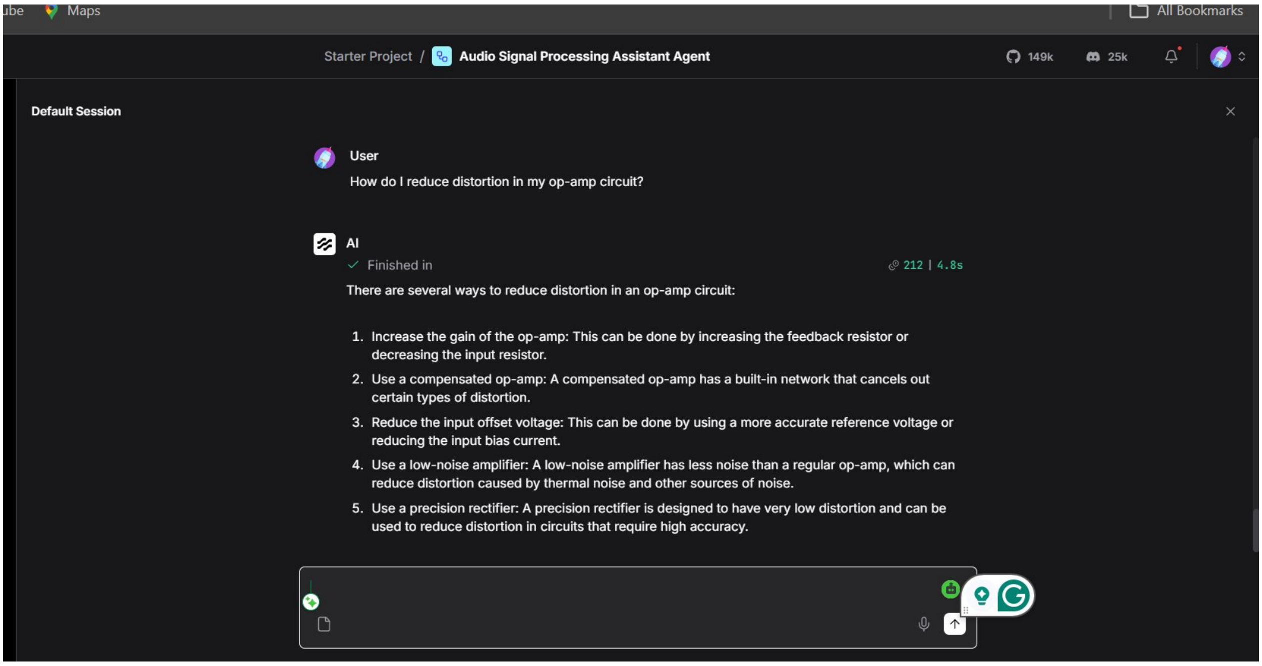


**Figure 3: Agent Response — Op-Amp Distortion Reduction**

Query: "How do I reduce distortion in my op-amp circuit?" The agent provided five structured engineering solutions including gain adjustment, compensated op-amp selection, input offset voltage reduction, low-noise amplifier substitution, and precision rectifier use.

## 8. IBM watsonx.ai Platform Integration

The project is deployed as a named project within IBM watsonx.ai — **"Audio Signal Processing Assistant Agent"**. The platform provides authentication, model routing, and API management for all LangFlow interactions. The dashboard below confirms active integration and token consumption during development and testing.

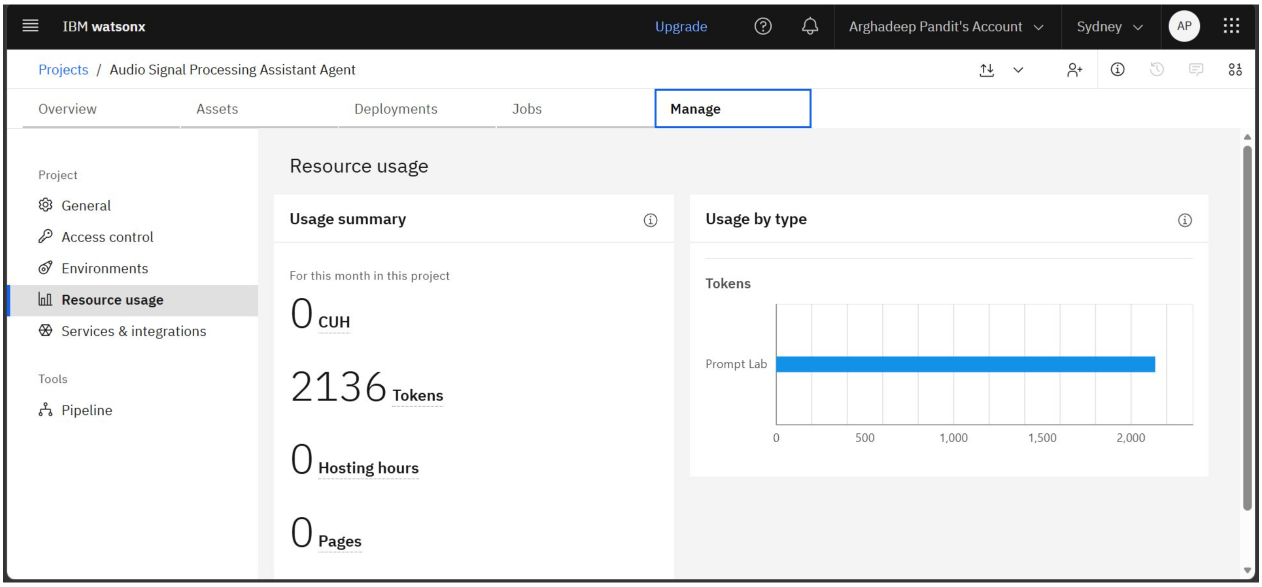


**Figure 4: IBM watsonx.ai Project Dashboard — Resource Usage & Token Tracking**

The Manage tab of the project dashboard shows 2,136 tokens consumed via the Prompt Lab during development and testing, confirming successful API integration with IBM Granite 8B. The watsonx API endpoint is configured at au-syd.ml.cloud.ibm.com.

## 9. Novelty & Uniqueness

| Feature | Distinguishing Factor |
|---|---|
| AI-Powered Audio Expert | Provides instant, interactive, context-aware guidance — unlike static documentation or generic chatbots. |
| Specialised Domain Knowledge | Purpose-built for audio engineering: filtering, sampling, quantization, noise, distortion, FFT. |
| Natural Language Interaction | Users describe problems conversationally — no technical query syntax or manual searching required. |
| Intelligent Troubleshooting | Analyses user-reported symptoms and suggests root causes, diagnostic steps, and corrective actions. |
| Dual-Purpose Support | Serves both educational (students) and professional (engineers) use cases within a single interface. |
| IBM Granite Integration | Leverages advanced enterprise foundation models for sophisticated domain reasoning and generation. |
| Low-Code Development | Demonstrates how sophisticated AI assistants can be built via LangFlow's visual workflow designer. |

## 10. Current Implementation Status

| Component | Status | Notes |
| --- | --- | --- |
| System Architecture Design | ✓ Completed | Core pipeline defined and validated |
| LangFlow Workflow Configuration | ✓ Completed | All five components connected and tested |
| IBM Granite 8B Integration | ✓ Completed | API authenticated; 2,136 tokens consumed |
| Prompt Engineering for Audio DSP | ✓ Completed | Domain-specific persona and constraints applied |
| Initial Prototype Demonstration | ✓ Completed | Multiple query types validated in Playground |
| Real-time Audio File Analysis | → Planned | Upload and auto-analyse .wav / .mp3 files |
| Spectrum Visualisation | → Planned | FFT-based interactive frequency plots |
| Voice Interaction | → Planned | STT / TTS for hands-free operation |
| IoT Audio Device Integration | → Planned | Smart device monitoring and diagnostics |

## 11. Future Scope

**Real-Time Audio File Analysis**

Future versions will allow users to upload .wav and .mp3 audio files for automatic analysis of noise, distortion, clipping, frequency response, and signal quality issues using integrated DSP libraries.

**Audio Spectrum Visualisation**

The system can be enhanced with waveform and spectrum analysis tools to provide graphical FFT representations and interactive frequency-domain dashboards.

**Voice-Based Interaction**

Integration of Speech-to-Text and Text-to-Speech technologies will enable fully hands-free operation — users speak their queries and receive spoken responses.

**Advanced DSP Design Assistance**

The assistant can be extended to provide filter design recommendations, equalizer settings, signal conditioning techniques, and DSP algorithm guidance for engineering applications.

**IoT Audio Device Integration**

The system can connect with microphones, speakers, smart audio systems, and IoT devices for real-time monitoring, diagnostics, and anomaly detection.

## 12. Applications & Expected Impact

| Use Case | Target User | Description |
| --- | --- | --- |
| Classroom Support | Students | Instant answers to audio DSP questions during study or lab sessions |

| Use Case | Target User | Description |
|---|---|---|
| Equipment Troubleshooting | Engineers | Quick diagnosis of microphone, speaker, amplifier, and interface issues |
| Signal Processing Learning | Students / Researchers | Interactive explanations of complex DSP theory and algorithms |
| Professional Support | Audio Engineers | Context-aware guidance for real-world signal processing problems |
| Equipment Design | Engineers / Designers | Filter selection, system optimisation, and component recommendations |

## 13. Conclusion

The **Audio Signal Processing Assistant Agent** represents a meaningful application of agentic AI to the audio engineering domain. By combining IBM Granite's advanced reasoning capabilities with specialised prompt engineering and LangFlow's low-code workflow orchestration, the system delivers an intelligent, accessible solution to real challenges faced by students and professionals alike.

This project demonstrates:

- Effective use of **agentic AI** for domain-specific, context-aware problem solving.
- Successful integration of IBM Granite **enterprise foundation models** in a practical application.
- The viability of a **low-code approach** to building sophisticated AI assistants.
- Significant **educational and professional value** in the audio engineering and DSP community.

With planned enhancements including real-time audio analysis, spectrum visualisation, voice interaction, and IoT integration, this assistant has strong potential for broader impact in audio education and professional engineering support.

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